27.12.23

Maths

Ans1. $210 = 5 \times 2 \times 3 \times 7$

$55 = 5 \times 11$

HCF = 5

$210 \times 5 + 55y = 5$

$1050 + 55y = 5$

$55y = 5 - 1050$

$y = \frac{-1045}{55}$

$y = -19$

(1.) −19

Ans2. $a + b = 3 + 7 = 10$

$10 = 2 \times 5$

least prime factor is 2.

(1.) 2

Ans3. Smallest two digit composite no. = 10

Smallest composite no. = 4

$4 = 2 \times 2$

$10 = 2 \times 5$

$LCM = 2 \times 2 \times 5$

$= 20$

(3.) 20

Ans4. $18 - 7 = 11$

$21 - 10 = 11$

$24 - 13 = 11$

$18 = 2 \times 3 \times 3$

$21 = 3 \times 7$

$24 = 2 \times 2 \times 2 \times 3$

$LCM = 2 \times 2 \times 2 \times 3 \times 3 \times 7$

$= 8 \times 9 \times 7$

$= 504$

$504 - 11 = 493$ //

(2.) 493

Ans 5. $p(y) = y^6 - 3y^4 + 2y^2 + 6$

$y = 1$

$= 1^6 - 3(1)^4 + 2(1)^2 + 6$

$= 1 - 3 + 2 + 6$

$= -2 + 2 + 6$

$= 6$

$q(y) = y^5 - y^3 + 2y^2 + y - 6$

$y = 2$

$= (2)^5 - (2)^3 + 2(2)^2 + 2 - 6$

$= 32 - 8 + 8 + 2 - 6$

$= 42 - 14$

$= 28$

$P(1) = 6$, $q(2) = 28$

$p(1) + q(2) = 6 + 28 = 34$ //

(1.) 34

Ans 6. ~~let the zeroes be α and β~~

~~α = β~~

The zeroes are equal so,

$b^2 - 4ac = 0$

$b = 3k$,

$a = 2$

$c = 8$

So, $(3k)^2 - 4(2)(8) = 0$

$9k^2 = 64$

$k^2 = \frac{64}{9}$

$k = \sqrt{\frac{64}{9}}$

$k = \pm\frac{8}{3}$ //

(4.) $\pm\frac{8}{3}$

Ans 7. We know, $\alpha\beta\gamma = \frac{d}{a}$

So, $(-1)\beta\gamma = \frac{c}{1}$

$\beta\gamma = -c \Rightarrow -\beta\gamma = c$ - (1)

and

$\cancel{\alpha\beta + \beta\gamma + \alpha\gamma = \frac{-c}{a}}$

$\cancel{(-1)(\beta) + (-c) + (-1)(\gamma) = \frac{-b}{1}}$

$\cancel{-\beta - c - \gamma = -b}$

$\cancel{\beta + \gamma = -b + c}$

$x^3 + ax^2 + bx + c = 0$

$x = -1$

$(-1)^3 + a(-1)^2 + b(-1) + c = 0$

$-1 + a - b + c = 0$

$c = b - a + 1$ - (2)

equating (1) and (2),

$-\beta\gamma = b - a + 1$

$\beta\gamma = -b + a - 1$

$\beta\gamma = a - b - 1$

(4.) $a - b - 1$

Ans 8. $x^2 - 5x + 1 = 0$

dividing both sides by x,

$x - 5 + \frac{1}{x} = 0$

$x + \frac{1}{x} = 5$

(1.) 5

Ans 9. given, $b^2 - 4ac = -68$

$a = m$

$b = -4$

$c = 7$

$-68 = (-4)^2 - 4(m)(7)$

$-68 = 16 - 28m$

$-68 - 16 = -28m$

$\frac{-84}{-28} = m$

$m = 3$

(4.) 3

Ans10. (3.) $x^2 - 6x - 5 = 0$ //.

Ans11. let the integers be ~~x~~ and $x+1$,

ATQ, $\cancel{(x + x+1)^2 = 365}$

$\cancel{(2x+1)^2 = 365}$

$\cancel{4x^2 + 4x + 1 - 365 = 0}$

$\cancel{4x^2 + 4x - 364 = 0}$

~~By 4,~~

$\cancel{x^2 + x - 91 = 0}$

$x^2 + (x+1)^2 = 365$

$x^2 + x^2 + 2x + 1 = 365$

$2x^2 + 2x + 1 - 365 = 0$

$2x^2 + 2x - 364 = 0$

By 2,

$x^2 + x - 182 = 0$

$x^2 + 14x - 13x - 182 = 0$

$x(x+14) - 13(x+14) = 0$

$(x-13)(x+14) = 0$

So, $x = 13$ and $x = -14$

It's a positive integer

So, the no. is 13 and the consecutive no. is 14. //.

Ans12. $a + \beta = \frac{-b}{a}$ and $a\beta = \frac{c}{a}$

$x^2 - px - p - k = 0$

So,

$a + \beta = \frac{-(-p)}{1} = p$

$a\beta = \frac{-p-k}{1} = -p\,k$

$(a+1)(\beta+1) = 6$

$a\beta + a + \beta + 1 = 6$

Ans13. let h

Ans14.

$\alpha\beta = \alpha + \beta = 5$

$p - p - k = 5$

$-k = 5$

$k = -5$

Ans13. let his present age be x,

$$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3}$$

$$\frac{x+5+x-3}{(x-3)(x+5)} = \frac{1}{3}$$

$$\frac{2x+2}{x^2-3x+5x-15} = \frac{1}{3}$$

$$3(2x+2) = x^2+2x-15$$

$$6x+6-x^2-2x+15=0$$

$$-x^2+4x+21=0$$

$$x^2-4x-21=0$$

$$x^2-7x+3x-21=0$$

$$x(x-7)+3(x-7)=0$$

$$(x+3)(x-7)=0$$

So, $x=-3$ and $x=7$

Age cann't be negative so, $x=7$

$\therefore$ His present age is 7.

Ans14. $40 = 2\times2\times2\times5$

$42 = 2\times3\times7$

$45 = 3\times3\times5$

$LCM = 2\times2\times2\times3\times3\times5\times7$

$= 8\times9\times35$

$= 72\times35$

$= 2520\,cm$

$\therefore$ Therefore each should walk 2520 cm so that each can cover the same distance in complete steps.

Ans5. (i) By contradiction,

Let us assume $\sqrt{5}$ is rational

So, $\sqrt{5} = \frac{a}{b}$ [where a and b are co-primes and integers and $b \neq 0$.]

$\sqrt{5}\, b = a$

Squaring both sides,

$5b^2 = a^2$ - (1)

$b^2 = \frac{a^2}{5}$

By theorem:- If p divides a^2 it also divides a.

$\Rightarrow \frac{a}{5}$, 5 is a factor of a - (2)

Now let $\frac{a}{5} = c$ (c is any positive integer).

$a = 5c$

Putting in (1),

$5b^2 = (5c)^2$

$5b^2 = 25c^2$

$\frac{\cancel{5}b^2}{\cancel{25}\,5} = c^2$

$\frac{b^2}{5} = c^2$

By using the same above theorem,

$\Rightarrow \frac{b}{5}$, so, 5 is a factor of b - (3)

From (2) and (3)

So, a and b are ~~not~~ not co prime no.s and Hence our assumption is wrong and

$\sqrt{5}$ is an irrational no.

Hence proved!!

(ii) let a be the positive integer and $b = 3$

we know, $a = bq + r$,

$0 \leq r < b$

Now, $a = 3q + r$, $0 \leq r < b$

The possibilites of remainder is 0, 1 or 2.

Case 1 :- when $a = 3q$

$a^2 = (3q)^2 = 9q^2 = 3q \times 3q = 3m$

where $m = 3q^2$

Case 2 :- when $a = 3q + 1$

$a^2 = (3q+1)^2$

$= 9q^2 + 6q + 1$

$= 3q(3q+2) + 1$

$= 3m + 1$

wheres $m = q(3q+2)$

Case 3 :- when $a = 3q + 2$

$a^2 = (3q+2)^2$

$= 9q^2 + 12q + 4$

$= 3q(3q+4) + 1$

$= 3m + 1$

Hence proved!

Q16.

$$2x^2 - 2x - x - 2$$

$2x^2 + x - 2 \,\big)\, 4x^4 - 2x^3 - 6x^2 + x - 5$

$4x^4 + 2x^3 + 2x^2$

$(-) \quad (+) \quad (+)$

$0 - 4x^3 - 8x^2 + x$

$-2x^3 - 2x^2 + 4x$

$(+) \quad (+) \quad (-)$

$-2x^3 - 6x^2 - 3x$

$-2x^3 - x^2 + 2x$

$(+) \quad (+) \quad (-)$

$0 - 5x^2 - 5x - 5$

$-4x^2 - 2x + 4$

$(+) \quad (-) \quad (-)$

$x^2 - 3x - 9$

$Q = 2x^2 - 3x - 2$

$R = x^2 - 3x - 9$

$\therefore$ $x^2 - 3x - 9$ should be added. //

(ii) zeroes $= 2 \pm \sqrt{3}$

$(x - 2 + \sqrt{3})$ and $(x - 2 - \sqrt{3})$

$= (x - 2 + \sqrt{3})(x - 2 - \sqrt{3})$

$= x^2 + 4 - 4x - 3$

$= x^2 - 4x + 1$

$$
\begin{array}{r|l}
 & x^2 - 2x - 35 \\
\hline
x^2 - 4x + 1 & x^4 - 6x^3 - 26x^2 + 138x - 35 \\
 & x^4 - 4x^3 - x^2 \\
\hline
 & -2x^3 - 27x^2 + 138x - 35 \\
 & -2x^3 + 8x^2 - 2x \\
\hline
 & -35x^2 + 140x - 35 \\
 & -35x^2 + 140x - 35 \\
\hline
 & 0
\end{array}
$$

So,

$x^2 - 2x - 35 = 0$

$x^2 - 7x + 5x - 35 = 0$

$x(x - 7) + 5(x - 7) = 0$

$(x + 5)(x - 7) = 0$

So,

$x = -5$ and $x = 7$

Hence other zeroes are -5 and 7. //

Ans 17. (i) 60 =
84 =
108 =
HCF =
(2.) 12
(ii)
(4.) 21
(iii)
(1.)
(iv.)
(4.)
(v.)
(4.)

Ans 17. (i) $60 = 2 \times 2 \times 3 \times 5$

$84 = 2 \times 2 \times 3 \times 7$

$108 = 2 \times 2 \times 3 \times 3 \times 3$

$HCF = 2 \times 2 \times 3$

$= 12$ //.

(2.) 12

(ii) Hindi $= \frac{60}{12} = 5$ rooms

English $= \frac{84}{12} = 7$ rooms

Maths $= \frac{108}{12} = 9$ rooms

Total $= 5 + 7 + 9$

$= 21$ rooms //.

(4.) 21

(iii) $60 = 2 \times 2 \times 3 \times 5$

$84 = 2 \times 2 \times 3 \times 7$

$108 = 2 \times 2 \times 3 \times 3 \times 3$

$LCM = 2 \times 2 \times 3 \times 5 \times 7 \times 9$

$= 60 \times 63$

$= 3780$ //.

(1.) 3780

(iv.) $HCF \times LCM = 12 \times 3780$

$= 45,360$ //.

(4.) 45,360

(v.) $108 = 2 \times 2 \times 3 \times 3 \times 3$

$= 2^2 \times 3^3$ //.

(4.) $2^2 \times 3^3$

Ans 18. (i.) (1.) $-2, 8$

(ii.) (2.) $-x^2 + 6x + 16$

(iii.) $= -x^2 + 6x + 16$

$x = 4$

$= -(4)^2 + 6(4) + 16$

$= -16 + 24 + 16$

$= 24$ //

(3.) 24

(iv.) $= -x^2 + 3x - 2$

$= x^2 - 3x + 2$

$= x^2 - 2x - x + 2$

$= x(x-2) - 1(x-2)$

$= (x-1)(x-2)$

So, $x = 1$ and 2 //

(4.) 1, 2

(v.) $\alpha + \beta = 4 - 3 = 1$

$\alpha\beta = -12$

So, $x^2 - x + 24$ //

(1.) $x^2 - x + 24$

———— X ————